

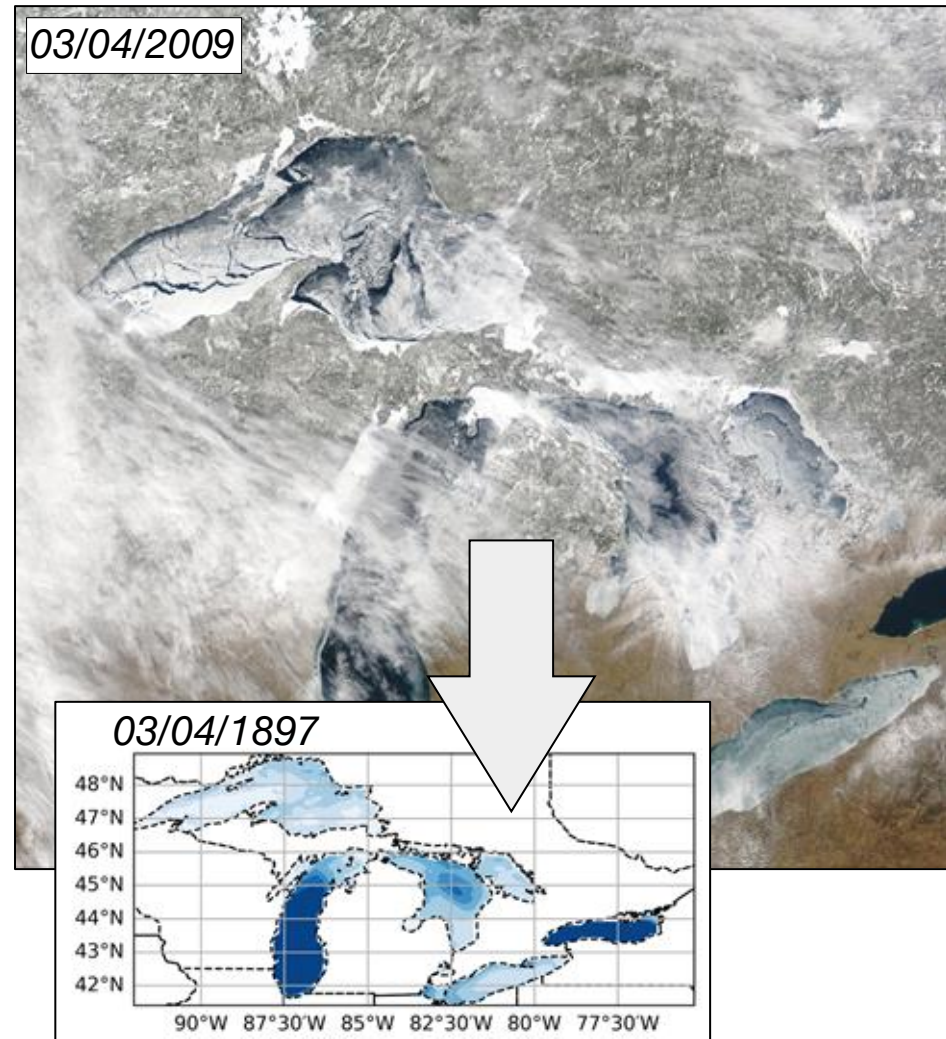
Deep Learning Approach for the Prediction and Historical Reconstruction of Spatial Ice Cover in the Laurentian Great Lakes

Jondall Norris

Great Lakes Summer Fellow
Cooperative Institute for Great Lakes Research,
University of Michigan

Mentors / Co-Authors:

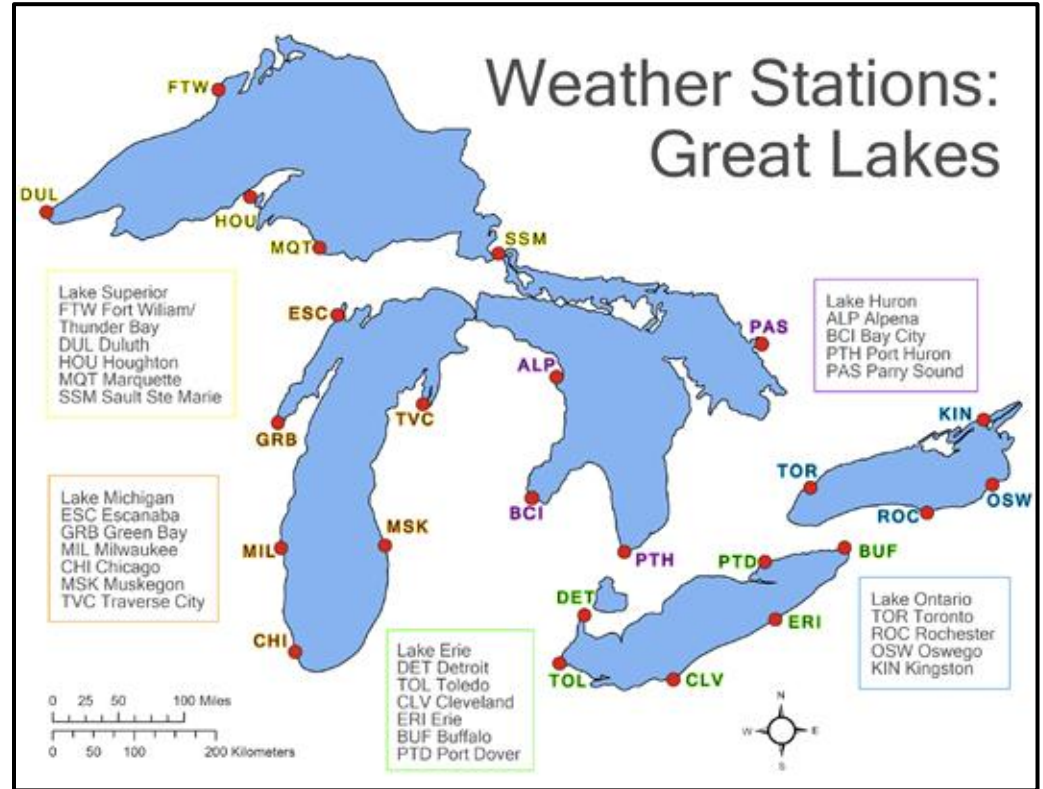
Hazem Abdelhady (CIGLR), David Cannon (CIGLR),
Ayumi Fujisaki-Manome (CIGLR),
James Kessler (GLERL), Dani Jones (CIGLR)



Motivation: Data and Precedent

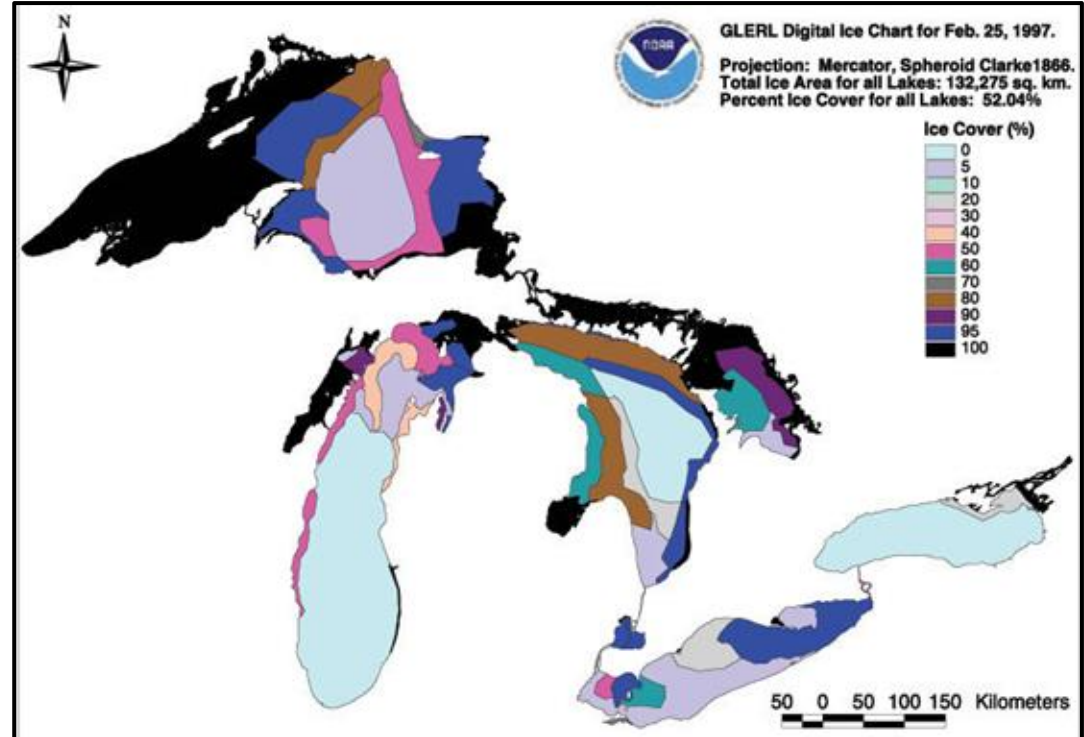
- Scattered air temperatures, 1897-2023

(Fujisaki-Manome et al., 2024)



Motivation: Data and Precedent

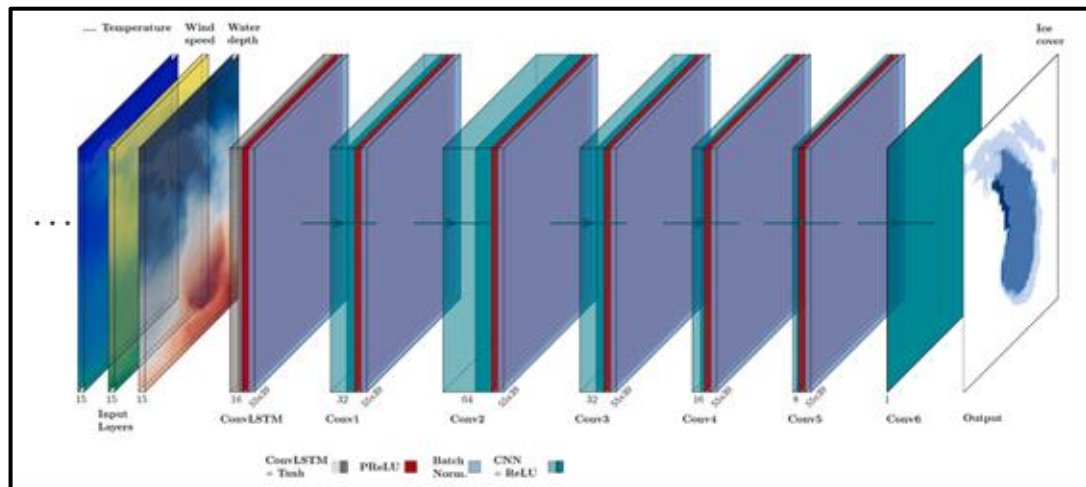
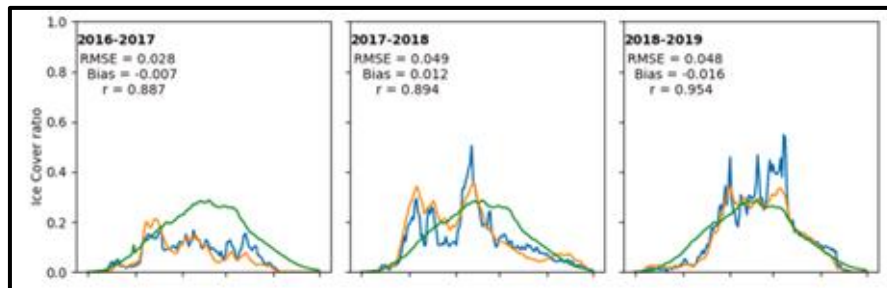
- Scattered air temperatures, 1897-2023
- Gridded ice concentrations, 1972-now (Yang et al., 2020)



Motivation: Data and Precedent

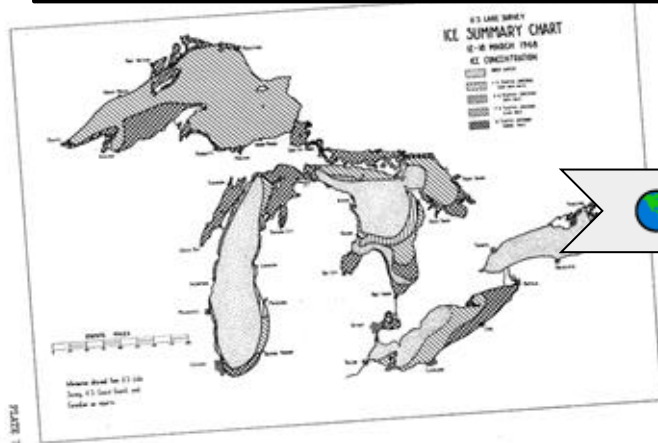
- Scattered air temperatures, 1897-2023
- Gridded ice concentrations, 1972-now
- **DL model of Lake Michigan ice cover**

(Abdelhady and Troy, 2024)



Motivation

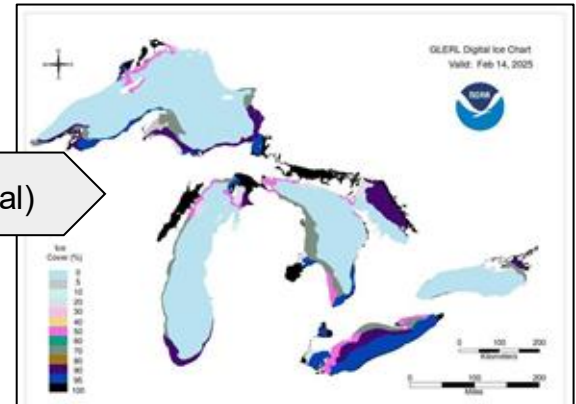
Can we predict Great Lakes ice cover, **spatially and temporally**, using only scattered historical air temperatures?



(sub-lake)



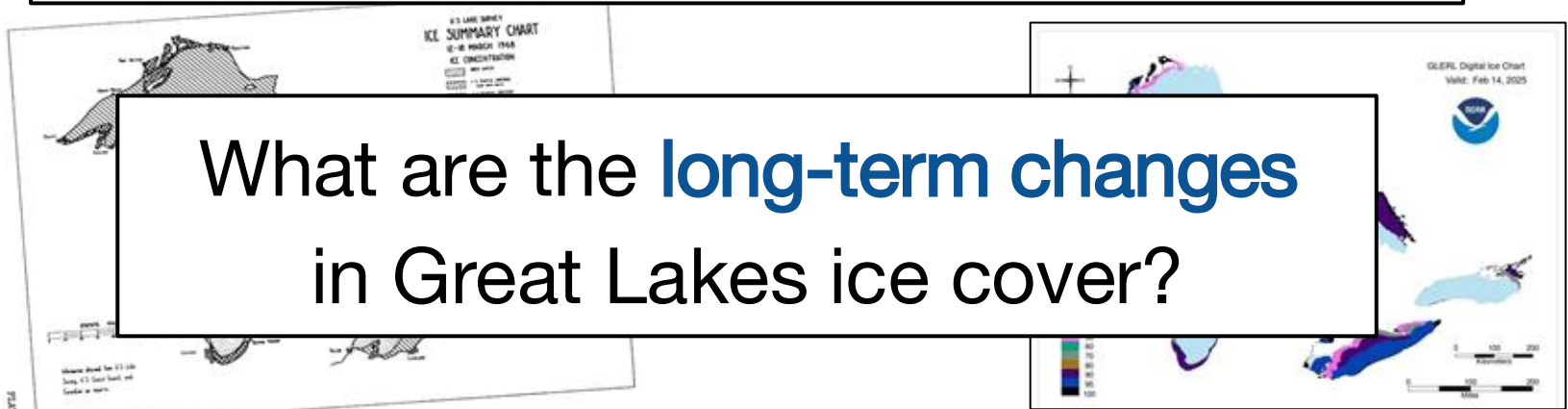
(sub-annual)



Motivation

Can we predict Great Lakes ice cover, **spatially and temporally**, using only scattered historical air temperatures?

What are the **long-term changes** in Great Lakes ice cover?



Developing a Deep Learning Model

Challenge 1:

How to get 2D gridded outputs over a lake from several nonuniform, land-based inputs?

Challenge 2:

How to make model learn underlying physical processes?

Challenge 3:

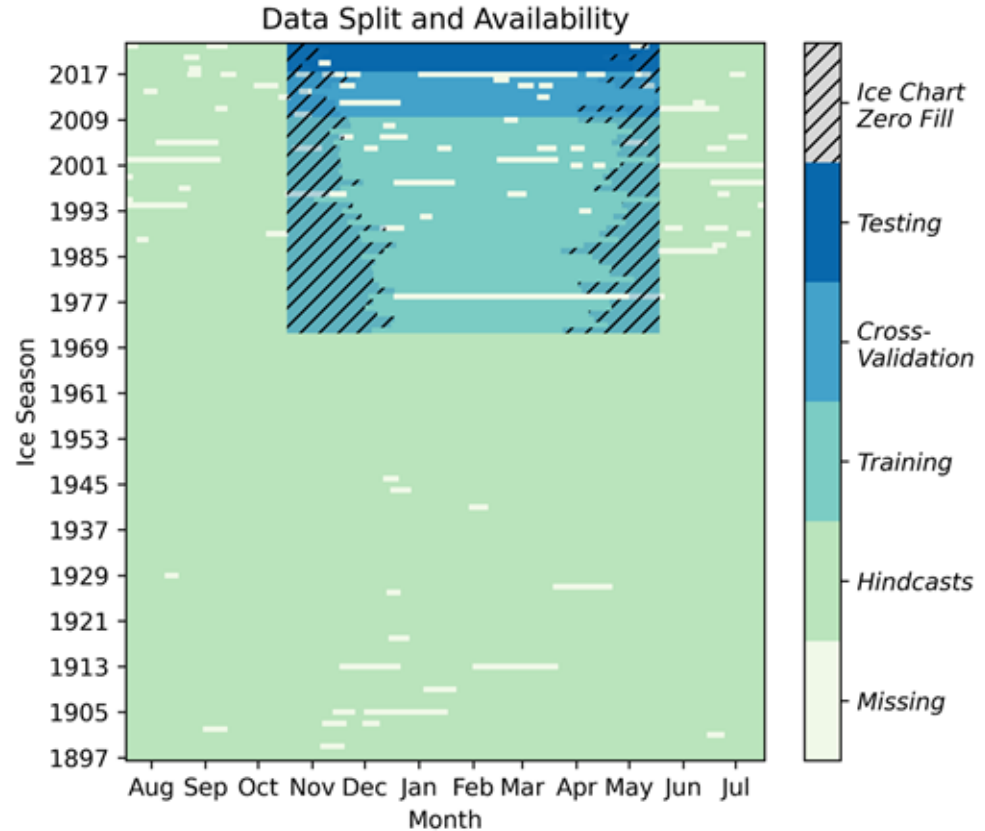
How to combine 1D inputs with 2D gridded inputs (e.g. depth)?

Challenge 4:

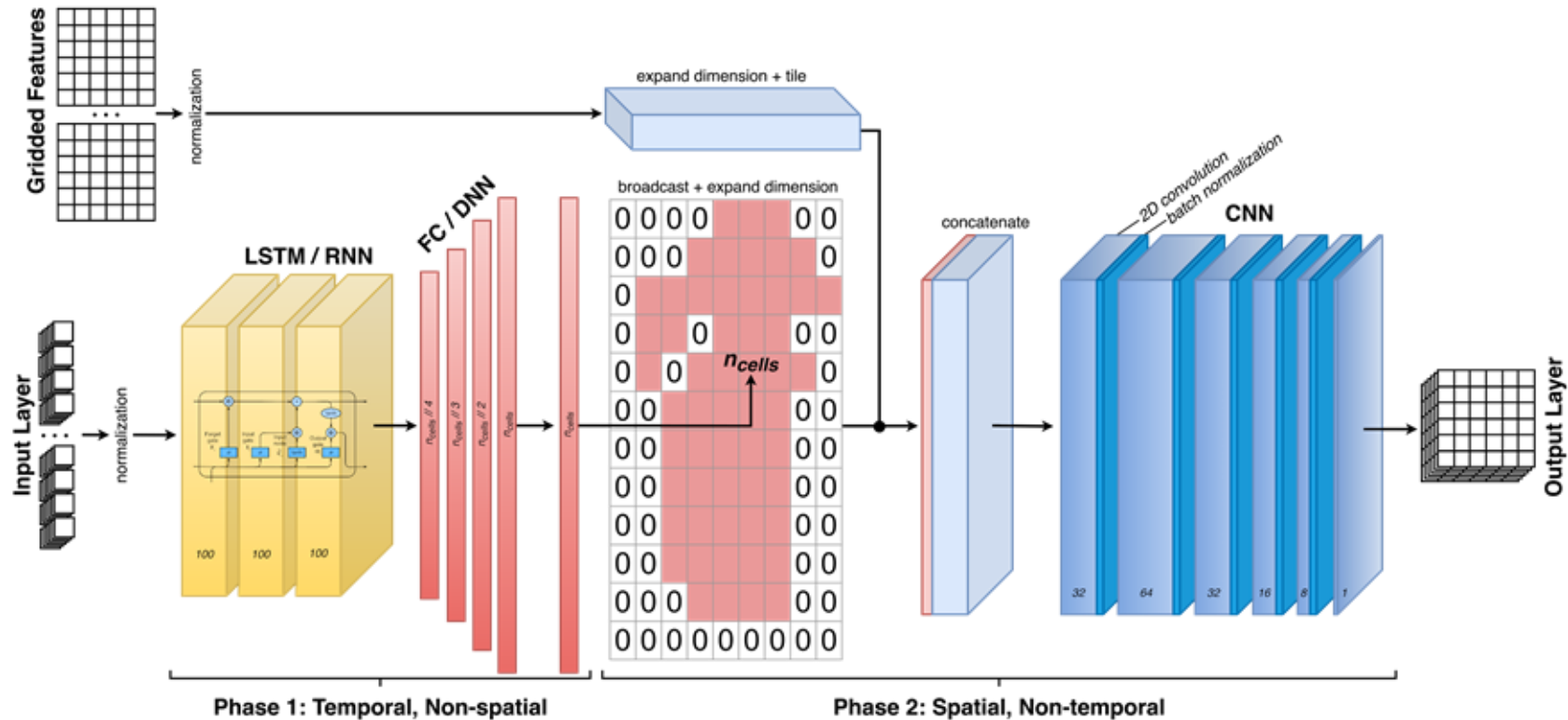
How to incorporate both spatial and temporal learning into one coherent model?

Model Architecture

- “Two-phase” design 🕒 ➡ 🌐
- Input and Training:
 - 🕒 daily average temperatures and trailing averages
 - 🌐 depth, shoreline prox., lat, lon
 - Ice charts to train and validate
- Output:
 - Daily ice concentration [%] at 18 km

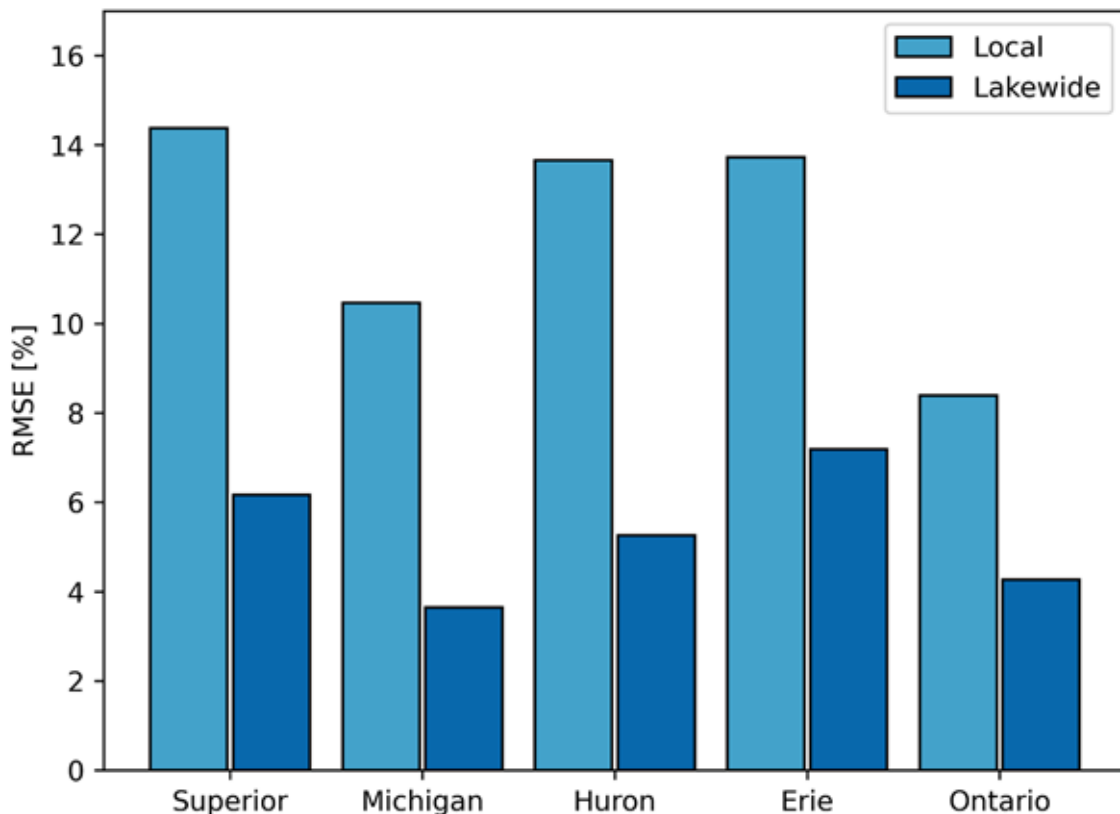


Model Architecture ()



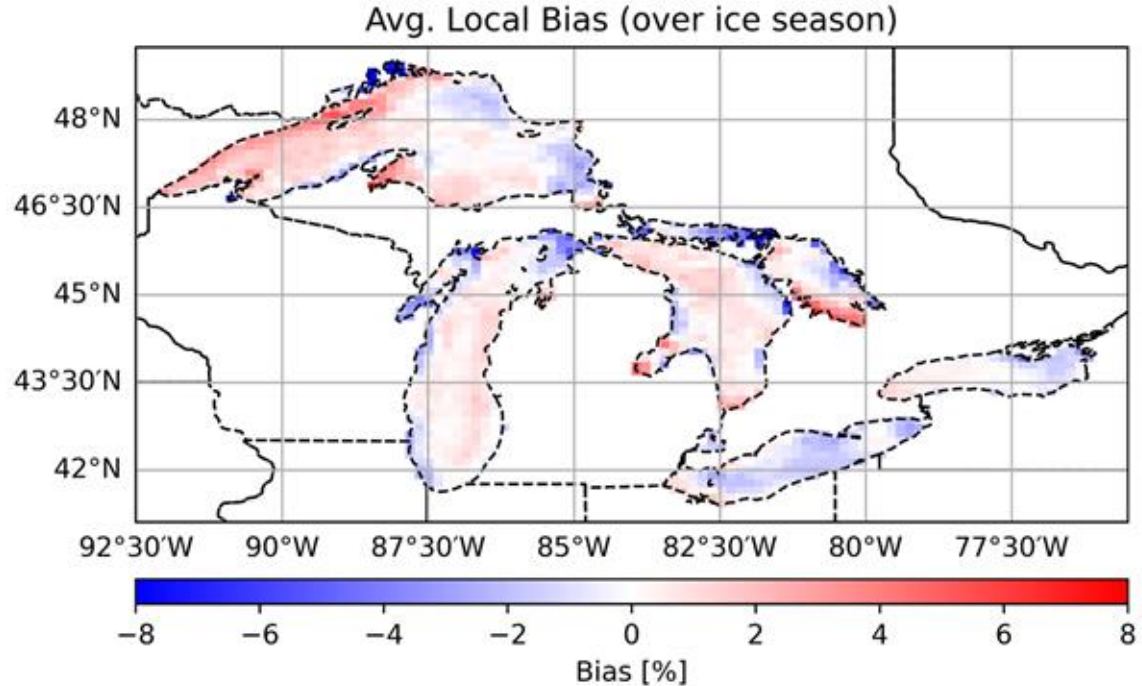
Model Performance

- 13 year validation
- Local:
8.4 - 14.4% RMSE
- Lakewide:
3.6 - 7.1% RMSE



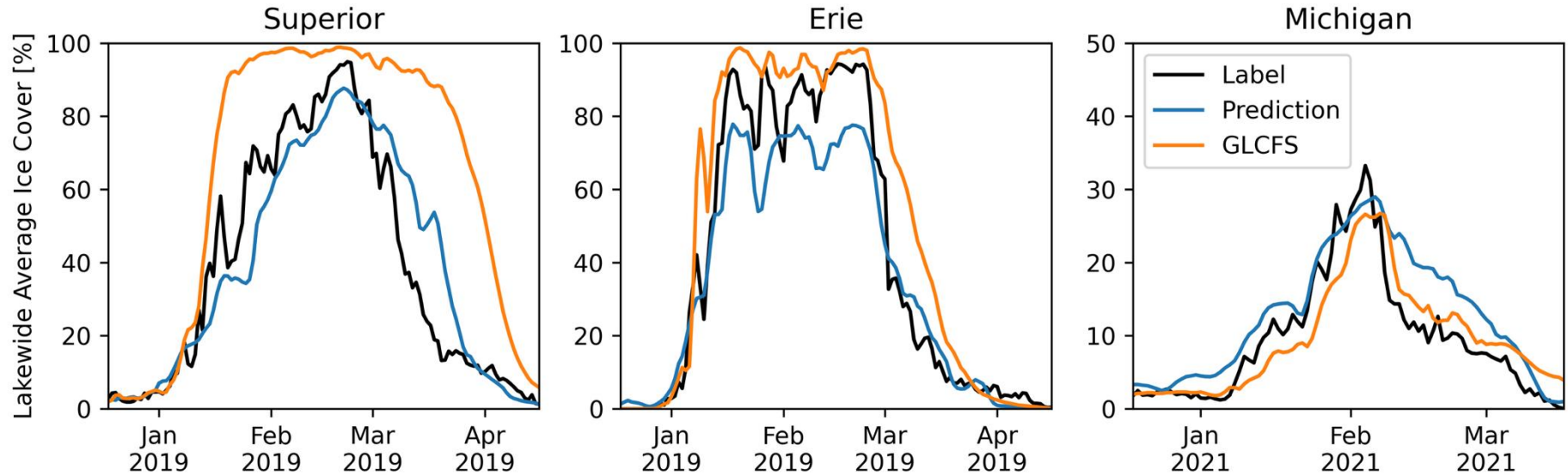
Model Performance

- 13 year validation
- Local:
8.4 - 14.4% RMSE
- Lakewide:
3.6 - 7.1% RMSE
- Spatial Error
 - Worse along coast and in bays/inlets



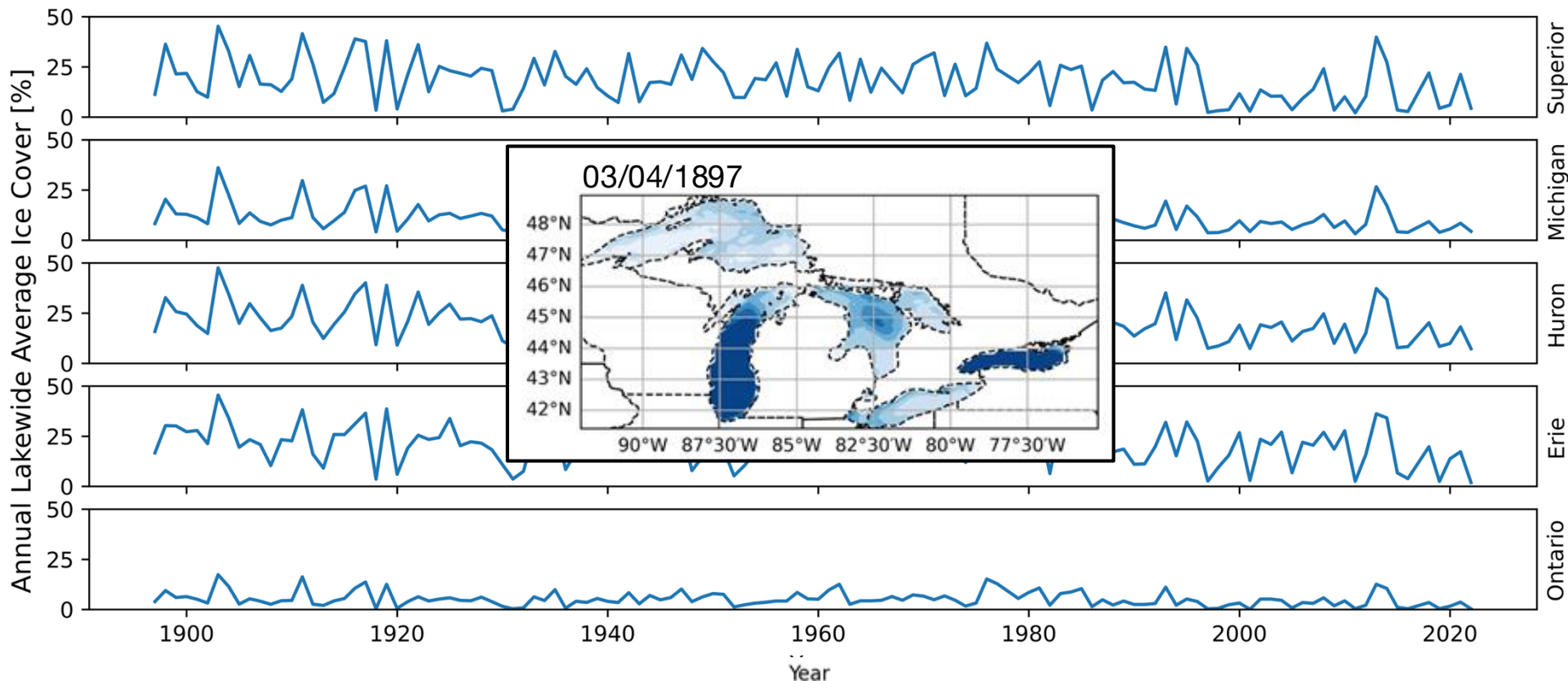
Performance against GLCFS

- Great Lakes Coastal Forecast System (FVCOM + CICE)

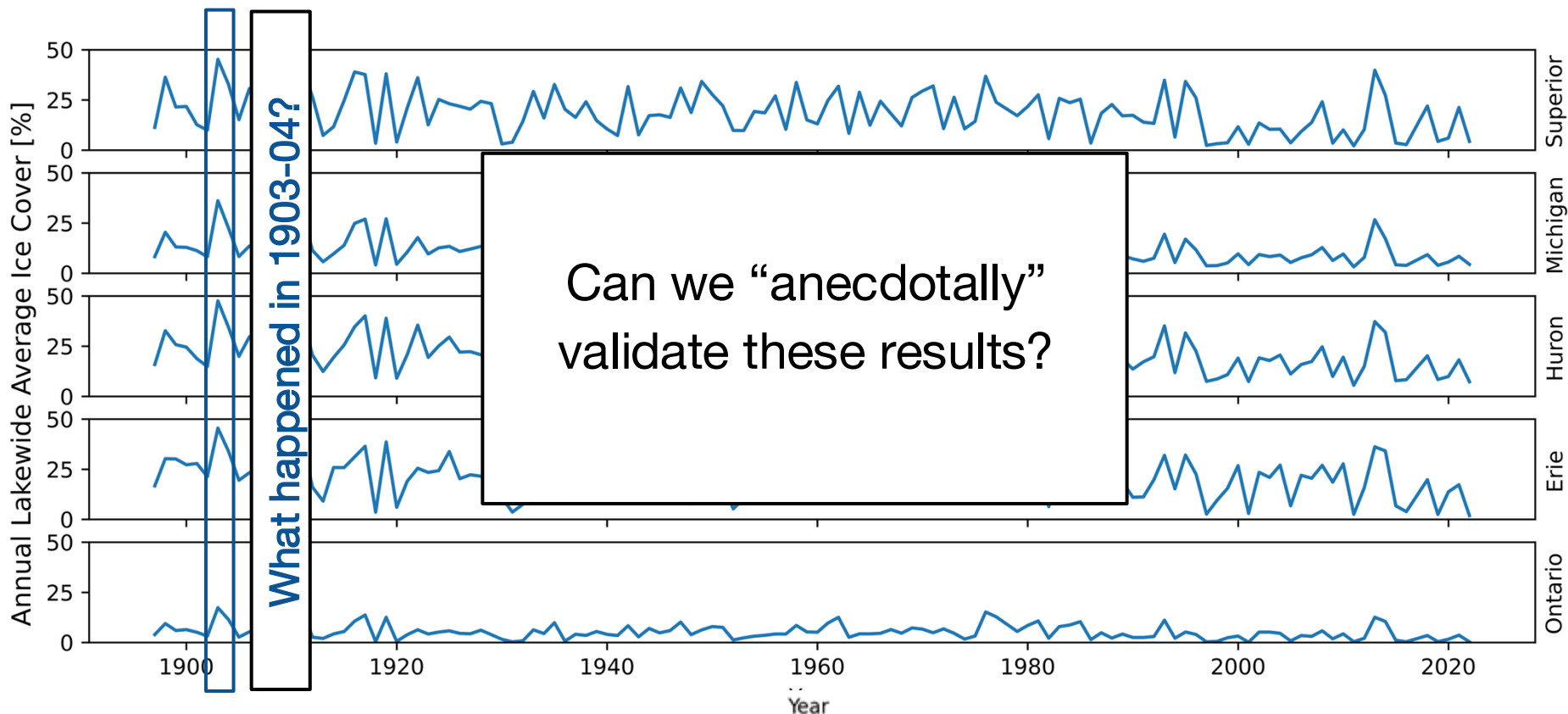


- Comparable performance (2019-2023): *within 1% or better RMSE*

Results: Apply Model to Historic Air Temperatures



Results: Apply Model to Historic Air Temperatures



Results: What Happened in 1903-04?

LAKE HURON.
No Ice Can Be Seen From Thunder Bay Island.
A PERILOUS TRIP.
Captain Persons Experienced It Saturday
—Today It Is a Wide Expanse of Solid Ice.

Capt. Persons says it is unprecedented to see Lake Huron frozen, so far out, so early in the season.

Alpena Evening News 12/14/1903

HANDIWORK OF ZERO
Ice Formations at Muskegon Attract People from Far and Near.

Berge More Than 100 Feet High.
Parties have pored on the ice in many different attitudes, and the pictures are being eagerly sought as souvenirs. The ice is the most extensive known here in many years, and the single bergs exceed in size those within the memory of the oldest inhabitant. Some of them are more than 100 feet in height and many exist which exceed 50 feet. The ice extended practically to the middle of Lake Michigan up to the time of the short thaw, when there was only a few miles of open water between the ice pack of this side and that of the west shore.

The True Northerner 02/12/1904

ICE FORTY FEET THICK.
Navigation at Milwaukee Suspended For First Time in Twelve Years.
Milwaukee, Wis., Feb. 16.—Navigation is suspended at this port for the first time in 12 years, ice conditions being the worst ever known to old lake men.

Alpena Evening News 02/16/1904

VERY UNUSUAL.
Teams Making Trips on the Ice From Bell to Middle Island.
The ice has not been sufficiently strong enough to carry a team to the island in many years.

Alpena Evening News 02/18/1904

Reports from Lake Superior are that there is more ice on that body of water than for many years past. A person could make his way over the ice from Marquette to Sault Ste. Marie, 180 miles down the lakeshore.

Alpena Evening News 02/27/1904

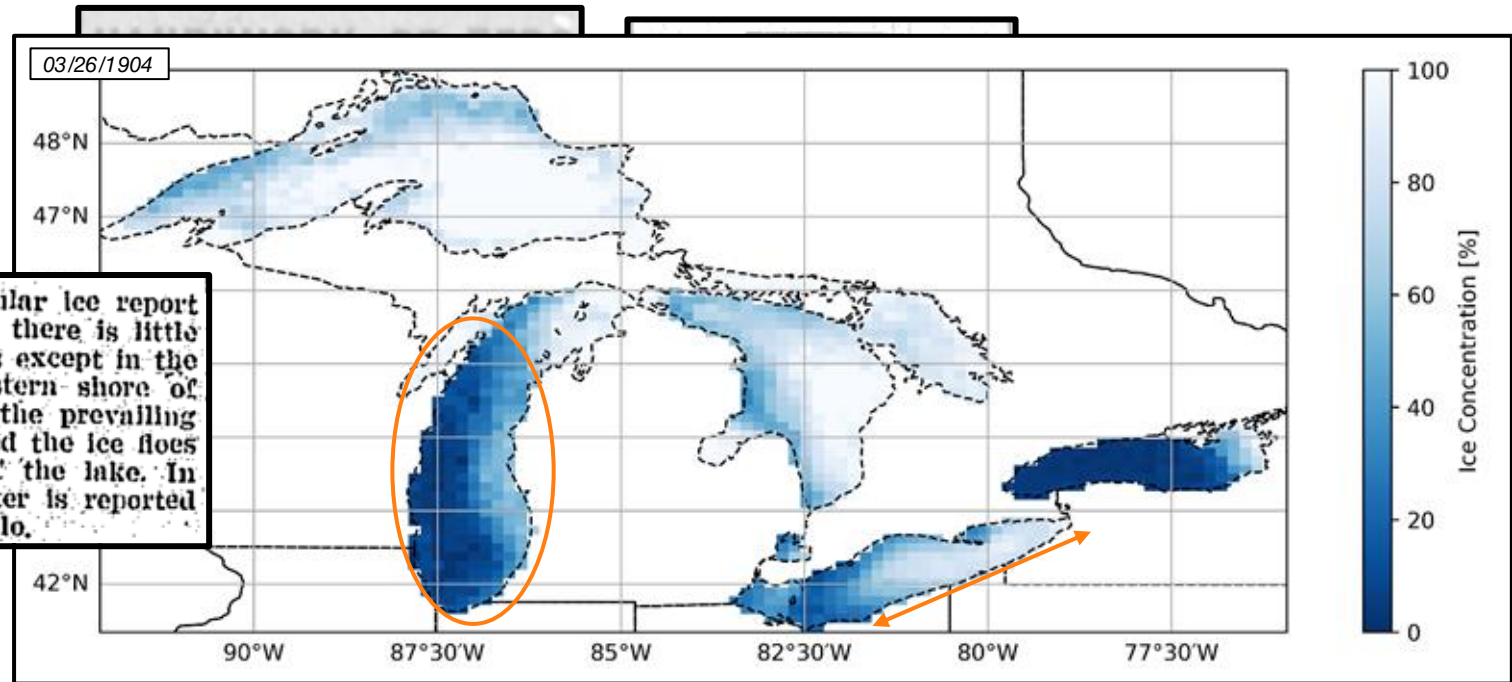
ICE WITHOUT END
eastern end the ice is solid. In comparison with same period last season there is more ice in all the lakes and the fields are more extensive.

Alpena Evening News 03/16/1904

According to the regular ice report of the weather bureau, there is little open water on the lakes except in the rivers and on the western shore of Lake Michigan, where the prevailing winds have again shifted the ice floes to the eastern shore of the lake. In Lake Erie no open water is reported from Cleveland to Buffalo.

Milford Times 03/26/1904

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water between the ice pack of this side and that of the west shore.

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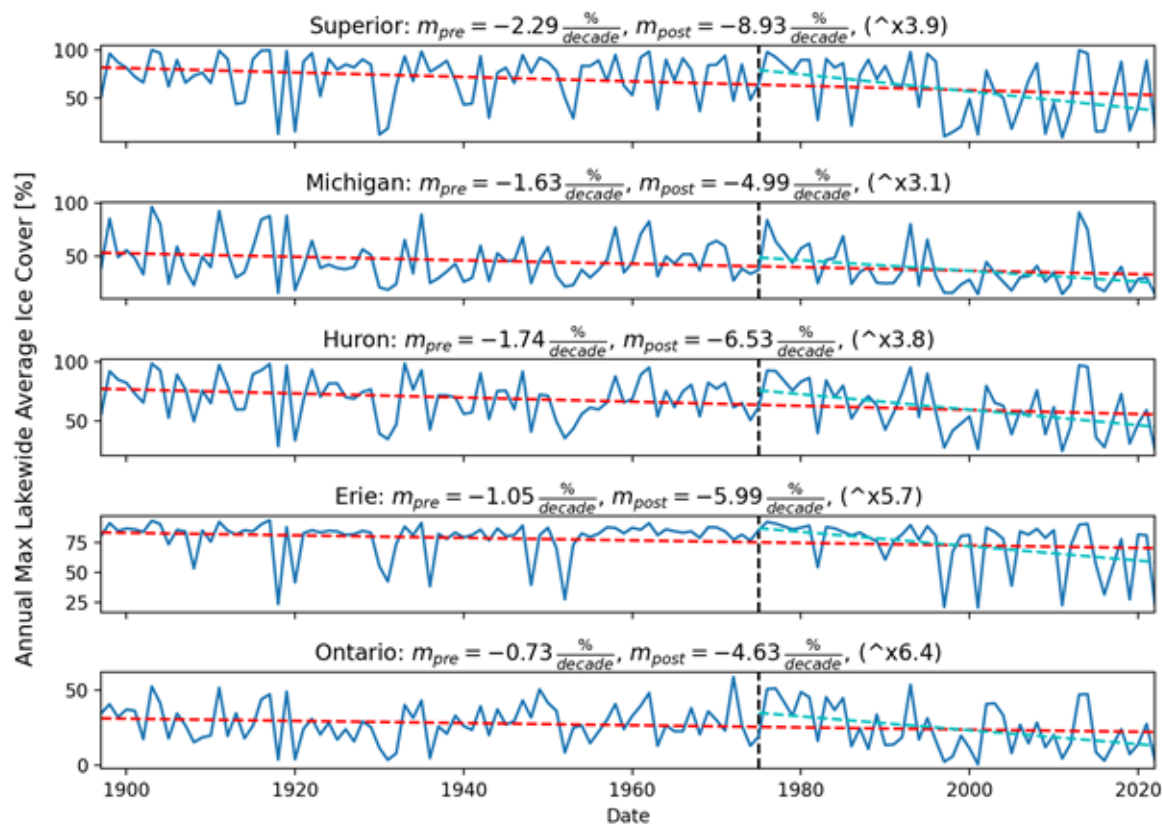
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Results: Trend Analysis

- Linear Trends in AAIC/AMIC
 - Now: -5 %/decade AMIC
 - 3-8x increase in ice loss trend post-1975



Conclusions

Sparse air temperatures can act as a proxy for ice concentrations in a **spatially and temporally resolved** DL model.

Great Lakes ice cover is **decreasing***

* Ice loss trends are weaker when reconstructed pre-satellite era:
Observational records must be contextualized.